

OMPRASAD BHASKAR PADWALKAR

2nd-Year Data Science Student & AI Developer

D Y Patil College of Engineering, Kolhapur, Maharashtra

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PROFESSIONAL SUMMARY

Highly motivated and results-driven 2nd-Year Engineering Student specializing in **Data Science, Machine Learning, and Artificial Intelligence** at D Y Patil College of Engineering, Kolhapur. Possesses a strong foundational understanding of statistical modeling, predictive analytics, computer vision pipelines, data structures, and full-stack web software engineering. Demonstrated technical expertise through the end-to-end development of real-world AI applications including financial transaction fraud detection systems, agricultural crop disease diagnosis models, and market analytics web solutions. Committed to leveraging data science methodologies and clean software engineering practices to solve complex real-world challenges.

EDUCATION & ACADEMIC BACKGROUND

Bachelor of Engineering (B.E.) in Data Science

2024 – Present (2nd Year)

D Y Patil College of Engineering, Kolhapur, Maharashtra (Affiliated to Shivaji University)

- Core Specialization:** Data Science Foundations, Artificial Intelligence, Machine Learning, Computer Vision, and Predictive Analytics.
- Key Coursework:** Data Structures & Algorithms (C/C++), Database Management Systems (SQL), Python Programming, Web Technologies, Applied Mathematics & Statistics.
- Academic Leadership:** Active member of departmental technical committees, leading peer learning sessions on Python programming, machine learning basics, and version control.

TECHNICAL SKILLS & CORE COMPETENCIES

- Programming Languages:** Python (Advanced), C, C++, JavaScript (ES6+), SQL, HTML5, CSS3
- Data Science & Machine Learning:** NumPy, Pandas, Scikit-Learn, OpenCV, Data Preprocessing, Feature Engineering, Anomaly Detection, Supervised & Unsupervised Learning
- Web Development & Frameworks:** HTML5, CSS3, JavaScript, Responsive Layout Design, RESTful APIs, AJAX, Swiper.js, Web3Forms, FormSubmit
- Developer Tools & Platforms:** Git, GitHub, VS Code, Linux Shell, Jupyter Notebooks, Google Colab
- Software Engineering Practices:** Content Security Policy (CSP) Hardening, Input Validation, Rate Limiting, Base64 Data Obfuscation, Dynamic PDF Generation
- Soft Skills & Professional Leadership:** Analytical Problem Solving, Strategic Team Leadership, Technical Documentation, Public Speaking, Interpersonal Communication

KEY FEATURED PROJECTS & PRACTICAL EXPERIENCE

SecurePay AI — Intelligent Payment Security Framework

AI / FinTech Security

- Project Overview:** Developed an intelligent financial payment security application designed to detect and prevent transaction fraud in real time.
- Technical Architecture:** Implemented machine learning anomaly detection models using Python and Scikit-Learn to evaluate transaction metadata and flag suspicious behavior.
- Key Impact:** Established real-time risk scoring metrics, user validation workflows, and automated threat mitigation to protect digital transaction integrity.

AgroScan AI — Agricultural Crop Disease Diagnostic System

Computer Vision / AgriTech

- **Project Overview:** Engineered a Computer Vision image diagnostic pipeline capable of identifying plant leaf diseases to assist farmers in early treatment.
- **Technical Architecture:** Utilized OpenCV for image pre-processing, contrast enhancement, color segmentation, and visual feature extraction from crop leaf images.
- **Key Impact:** Empowered local agricultural communities with rapid diagnostic feedback, actionable treatment recommendations, and yield preservation insights.

KrishiSetu — Bridge for Farmers (Data Analytics Platform)

Data Analytics / Full-Stack Web

- **Project Overview:** Built a comprehensive agricultural data analytics platform connecting farmers directly with transparent market pricing and buyer networks.
- **Technical Architecture:** Integrated market price analytics, regional crop data aggregation, dynamic UI visualization, and responsive mobile-first architecture.
- **Key Impact:** Eliminated middleman exploitation by providing real-time price trend forecasting and direct buyer connectivity for agricultural producers.

80th Independence Day Wishes Platform

Interactive Web Solution

- **Project Overview:** Created an engaging web application for generating personalized national celebration greetings with dynamic multimedia components.
- **Technical Architecture:** Developed custom greeting generation engines, background audio controls, keyframe CSS animations, and social share links.
- **Key Impact:** Achieved widespread user engagement through seamless social sharing capabilities and responsive design across all mobile and desktop devices.

Personal Developer & Data Science Portfolio

Full-Stack Web Engineering

- **Project Overview:** Architected and deployed a futuristic, highly responsive personal developer portfolio featuring automated GitHub project synchronization.
- **Technical Architecture:** Implemented strict Content Security Policy (CSP) meta headers, rate-limiting anti-spam controls, confidential contact form integration, and vector PDF resume export.
- **Key Impact:** Maintained 100% production uptime, optimized SEO indexing for search visibility, and ensured bulletproof security against cross-site scripting.

CERTIFICATIONS, HACKATHONS & PROFESSIONAL ACHIEVEMENTS

- **Hack2Skills Innovation & Hackathon Certificate:** Honored for rapid technological prototyping, cross-functional collaboration, and innovative solution design under competitive time constraints.
- **Python for Data Science Certification:** Completed comprehensive certification covering numerical computing with NumPy, data manipulation with Pandas, data visualization, and ML model building.
- **Soft Skills & Communication Training (Level 1):** Certified in effective workplace communication, technical presentation techniques, teamwork dynamics, and professional etiquette.
- **Advanced Soft Skills & Leadership Mastery (Level 2):** Awarded advanced credential in corporate readiness, team management, strategic decision-making, and emotional intelligence.
- **Technical Skill Development & Engineering Excellence:** Standardized assessment credential recognizing engineering problem-solving capabilities and project execution.

LEADERSHIP, MENTORSHIP & EXTRA-CURRICULAR ACTIVITIES

- **Technical Peer Mentor:** Conducted introductory sessions for junior engineering students on Python programming, version control with Git/GitHub, and data science fundamentals.
- **Event Coordinator:** Actively organized and coordinated technical events, coding competitions, and Data Science workshops at D Y Patil College of Engineering, Kolhapur.
- **Continuous Learning:** Passionate about exploring cutting-edge AI advancements, Large Language Models (LLMs), Computer Vision research, and open-source software contribution.